

CUSTOM FIRMWARE · SEEED HARDWARE · 3D-PRINTED CASE

Cairn

for the Seeed Wio Tracker L2 Pro

Off-grid messaging that works on its own. A LoRa mesh messenger with a touch interface built for the trail — no network, no account, nothing to pay.

Firmware, case and design by **Christopher Van Hoose** · [CVH VISUALS](#).com



MeshCore v1.17.1

CAIRN 106

6 tabs

TOUCH UI, ONE
THUMB

GNSS · Audio

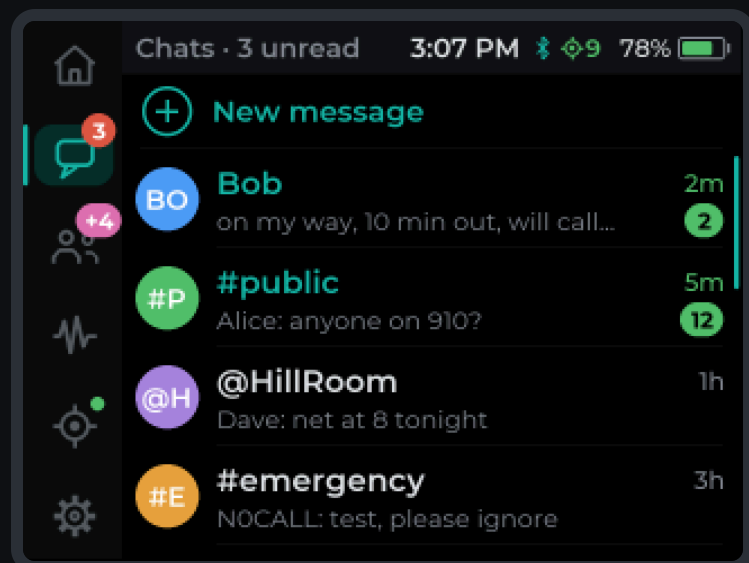
FIX, MAP, ALERT
TONES

3000 mAh

USB-C CHARGING

Everything happens on the device.

Cairn is a build of the open-source MeshCore companion firmware for Seeed's Wio Tracker L2 Pro, with a native touch interface. Read and reply, see who is around, check the radio and get a fix without reaching for a phone — and when you do want one, it still pairs with the official MeshCore apps over Bluetooth.



Chats. Channels and direct messages in one list, unread counts on the tab, a new-message button where your thumb already is.

HOME

Status at a glance: battery and time left, chats, contacts and the last node heard, GNSS, radio, Wi-Fi and Bluetooth.

CHATS

Channels and DMs, threads with route and time on incoming messages, quick replies, a full keyboard.

NODES

Who is on air and who you know — hops, last heard, distance and bearing.

RADIO

Live signal against the noise floor, the packet log, a full spectrum view and band scan, and an advert in two taps.

GPS

Position, altitude, speed and heading with a compass; a map; an Eco mode that powers the receiver down between fixes.

SETTINGS

Six groups: Radio, Connect, Display, Location, Controls, System — region and TX power, Bluetooth PIN, Wi-Fi, tones, maps, buttons, lock, updates.

Works alone

NO PHONE REQUIRED

Stays compatible

MESHCORE APPS
OVER BLE

Made for outside

GNSS, TONES, LOCK,
HIBERNATE

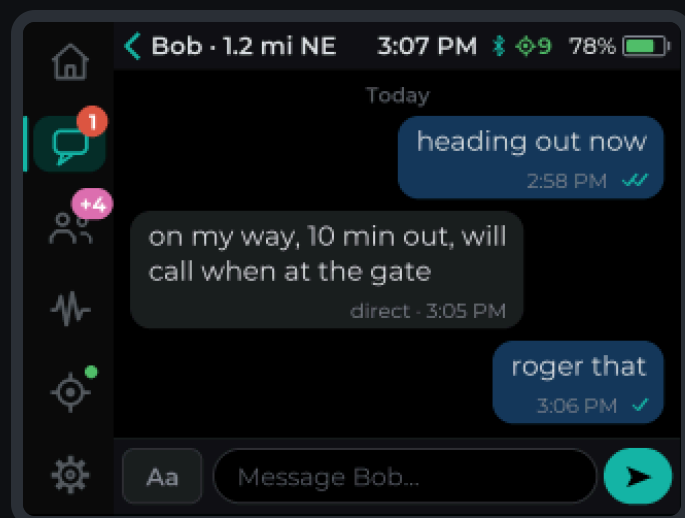
Actively developed

TRACKS UPSTREAM
MESHCORE

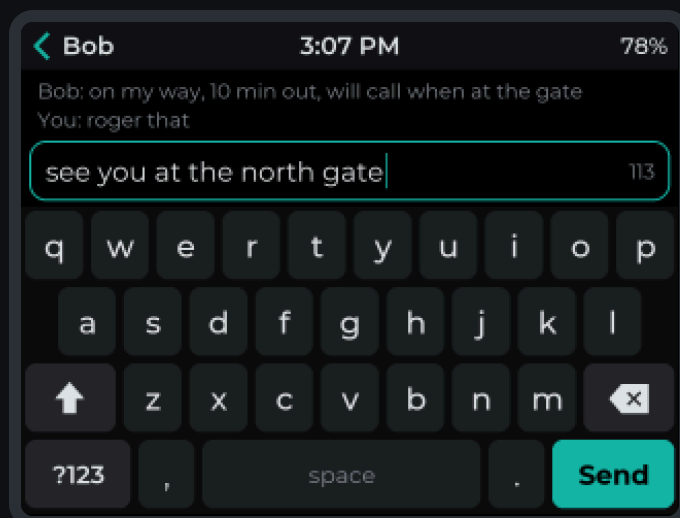
Physical where it matters.

Touch for reading and writing; real buttons for the things you do without looking. User/Boot opens the Quick panel, starts a new message or goes back (press, double-press, hold — each one reassignable in Settings). WAKE on the top edge turns the screen off, locks on a hold and unlocks on a double-press. Page 4 has the details.

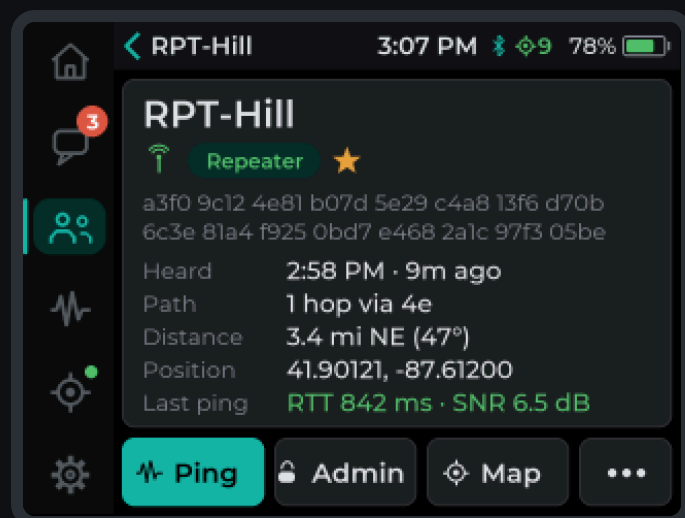
Read it, answer it, right there.



Thread. A conversation with time and route (direct or how many hops) on each incoming message, and a tick when yours has gone out. Preset quick replies send after one confirm.



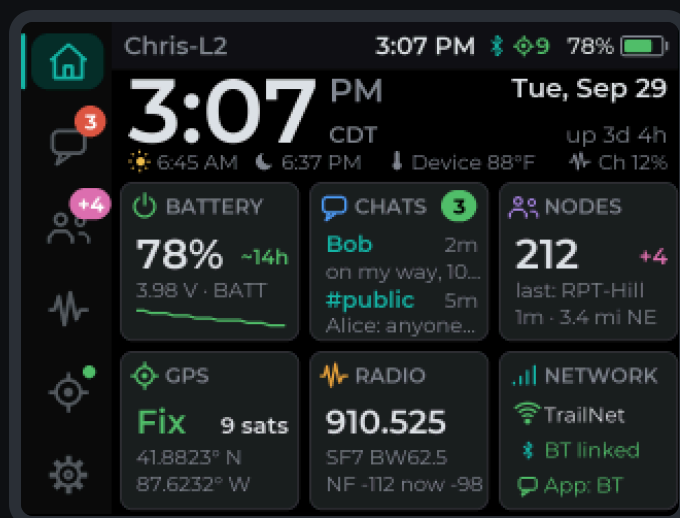
Keyboard. Write a full message on the touchscreen; a counter shows the bytes left as you near the limit (160 bytes for a DM, a little less in a channel).



Node detail. A node's key, when it was last heard, the path it came by, distance and bearing, position. Ping it, message it, put it on the map.

Messages that stay

Your last 200 messages, received and sent, survive a power cycle (chat history is kept by default and can be switched off), and what you send from the device shows in its own list — not just in the app.

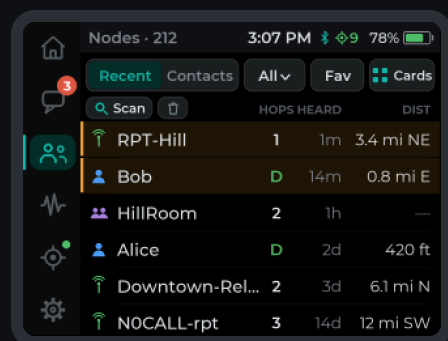


Home. The dashboard you wake to: battery with time remaining, chats, nodes, GNSS fix, radio setup, Wi-Fi and Bluetooth.

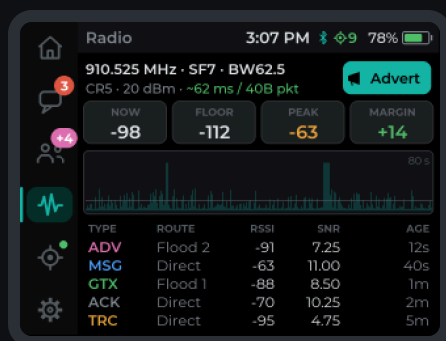
Rooms and channels

Public channels, room servers and direct messages all appear in Chats. A room password can be saved on the device so the room logs you back in automatically.

Know the mesh around you.



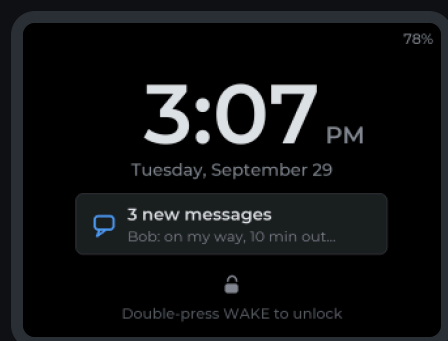
Nodes. One list, two sources: Recent (advertises heard on air) and Contacts (everything saved). Hops, last heard, distance and bearing; filter by type.



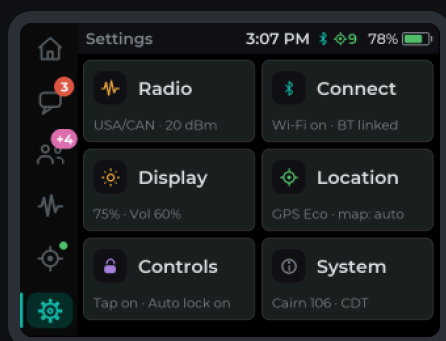
Radio. Frequency, SF, bandwidth, CR and power; NOW, FLOOR, PEAK and MARGIN; an 80 s signal history that opens the full Spectrum view; the packet log with type, route, RSSI, SNR and age. Advertise, then Send, to announce yourself.



GPS. Position, altitude, speed, heading on a compass, fix age and satellites. Map view. Eco mode powers the L76K down between fixes and tells you when the next one is.



Lock screen. A big clock and the latest message. Hold WAKE to lock; double-press WAKE or User/Boot to unlock. Touch is ignored while locked unless you turn on touch unlock.



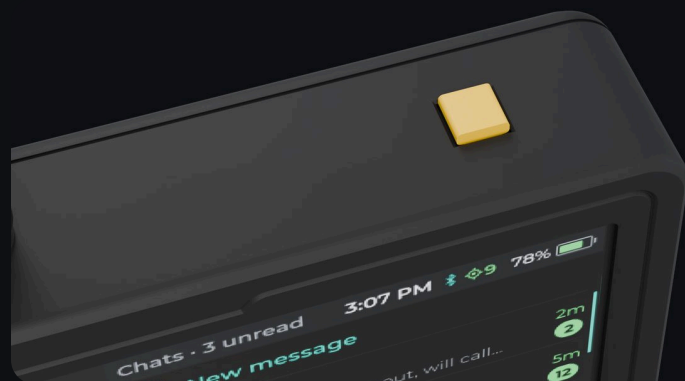
Settings. Six groups: Radio (region, TX power, channels), Connect (Bluetooth PIN, Wi-Fi), Display (display, sound), Location (GPS Off/On/Eco, maps), Controls (buttons, lock), System (clock, updates, hibernate).

Six tabs, one thumb

The tab bar never moves. On the engraved case, six marks beside the screen line up with it, so you find a page by feel before the screen is lit.

Buttons that help

User/Boot: press for the Quick panel, double-press for a new message, hold for Back — all reassignable; triple-press always mutes. Hold it during the boot splash for a USB recovery console.

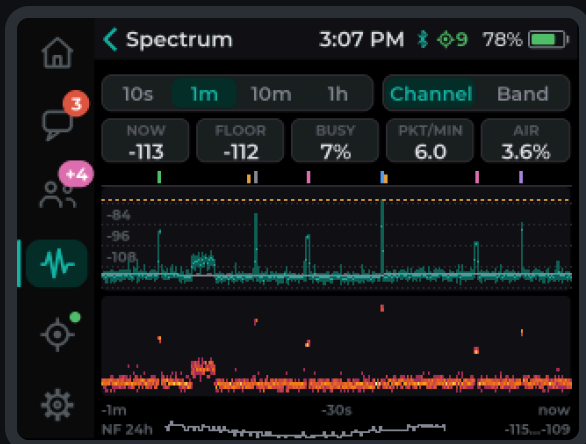


Wake it, lock it, leave it.

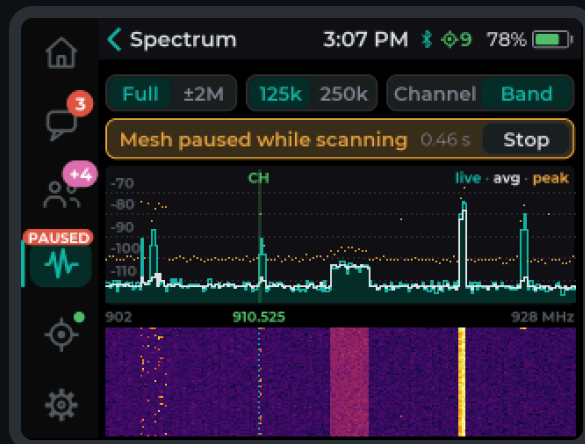
WAKE sits on the top edge where a thumb finds it in a pocket. Press to turn the screen off, hold to lock, double-press to unlock; with auto-lock on, every screen-off also locks. Hibernate (Settings → System) powers the device down for the long haul — User/Boot brings it back.

See the air, find the repeaters.

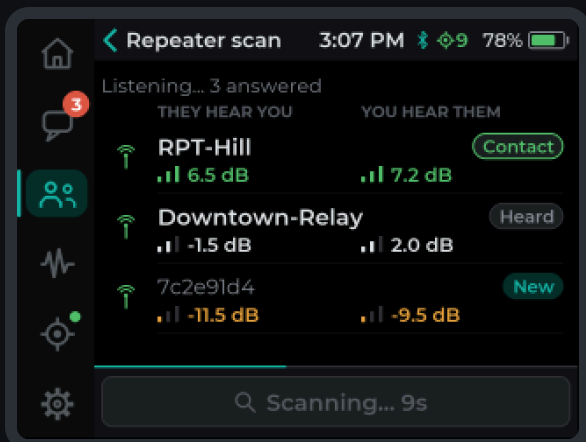
Check the channel, sweep the band, find the repeaters that hear you and look after your own — on the device, with no laptop or app.



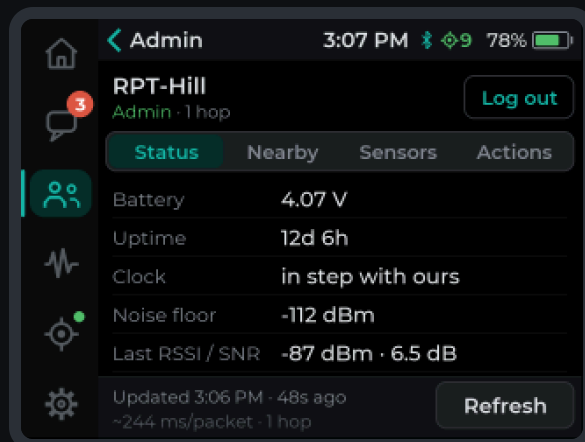
Spectrum. Your channel from 10 s to 1 h: busy time, packets per minute, airtime.



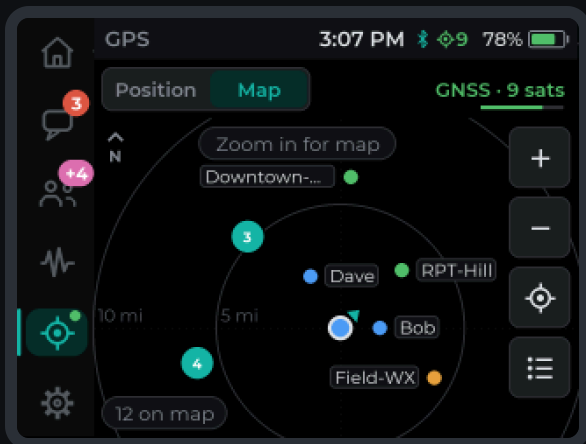
Band scan. Sweeps the whole band with a waterfall; the mesh pauses while it runs.



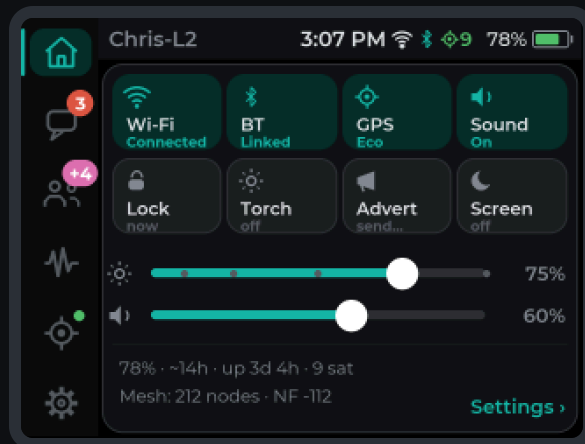
Repeater scan. Which repeaters in direct range hear you, and how well you hear them.



Repeater admin. Log in to your repeater: status, neighbours, sensors, safe actions.



Map. Nodes that share a position, on range rings or offline microSD maps.



Quick panel. Wi-Fi, Bluetooth, GPS, sound, lock and torch in one tap each; advert in two.

Engineering you can feel.

Small decisions that make a handheld radio feel polished.

RADIO AND MESH

- MeshCore v1.17.1 underneath: flood routing that learns direct paths, end-to-end encryption, repeaters and room servers.
- Spectrum samples only while the receiver is actually listening — never during the node's own transmission — so the trace is the channel, not the node.
- Receiver watchdog: the radio self-checks and resumes listening automatically.

POWER

- GPS Off and Eco fully remove power from the GNSS module between uses.
- Hibernate drops every rail and deep-sleeps; User/Boot wakes it.
- Brightness drives the backlight; the battery page shows USB power and time remaining from a self-scaling discharge log.

AUDIO

- Direct message, channel, sent, advert and startup tones on the ES8311 codec and the 0.8 W speaker; set them under Settings → Display → Sound, and triple-press User/Boot to mute.
- The audio clock is idle between tones to keep GNSS reception clean.
- A tone that interrupts another fades the old one out instead of clicking.

UPDATES

- Settings → System → Firmware: install a new release over Wi-Fi when you choose, or upload one from a computer's browser. Nothing installs by itself.
- Updates go to the spare app slot, so an interrupted upload never replaces the running firmware.
- USB flashing with a single full image. The build (BUILD 106 on the splash, Cairn 106 in Settings → System → About) shows which firmware is running.

RELIABILITY

- The most destructive actions — hibernate, factory reset, clearing chat history, rebooting a repeater — needs a deliberate hold, then a confirm.
- A 15 s watchdog restarts a frozen interface, and About shows why the last restart happened.

HERITAGE

- Developed first on the Wio Tracker L1 Pro, then brought to the L2 Pro.
- It tracks upstream MeshCore, so protocol fixes arrive with each merge.

Built on Seeed's Wio Tracker L2 Pro.



In the Cairn case: power slider, User/Boot, microSD, RST and the LED window in one strip on the right edge; WAKE on top.

THE BOARD · SEEED'S SPECIFICATIONS

Processor	ESP32-S3R8 (Wio-S3 module), Wi-Fi and Bluetooth LE
LoRa	Semtech SX1262, 863–928 MHz
Display	3.2" IPS, 320 × 240, capacitive touch
GNSS	Quectel L76K (GPS, BeiDou, GLONASS, QZSS), ceramic patch antenna
Audio	ES8311 codec, 6 Ω 0.8 W speaker, microphone
Storage	microSD card slot
Charging	USB-C 5 V; solar input on the board
Battery	3000 mAh LiPo
Expansion	Grove port beside the USB-C
Controls	Power switch, User/Boot, RST, WAKE; mesh and charging LEDs

Per Seeed Studio's Wio Tracker L2 Pro documentation.

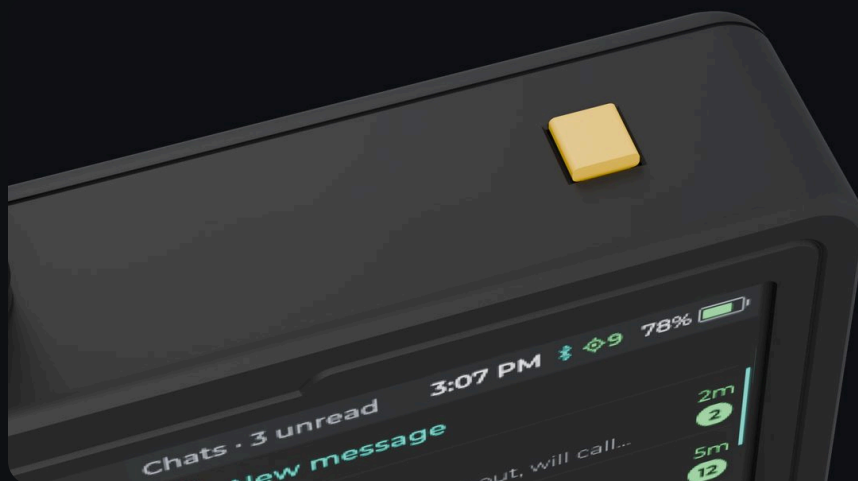
CAIRN CASE ADDITIONS · NOT SEEED SPECIFICATIONS

87.5 × 69 × 26.3 mm

CAIRN CASE, ANTENNA OFF

RP-SMA

STRAIGHT ANTENNA
REPLACES THE STOCK
FOLDING ONE



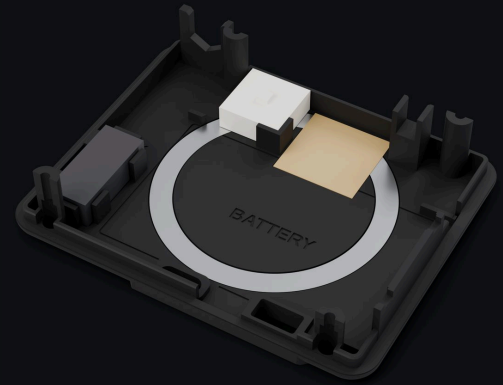
The top edge of the Cairn case: WAKE sits where a thumb finds it.

THE CASE

A moulded look, printed in PETG.



Built from the back. The front shell is bezel and walls in one piece; the board and parts go in from the open back, a flat plate closes it, and the buttons click in last from outside.



The open plate. An optional stick-on magnetic ring drops into its sunk seat; GNSS, speaker and Wi-Fi go where the floor map shows.



Closed-in ports. A USB-C pocket that takes any cable; a Grove grip pocket with a 45° funnel down to the plug.

Size	87.5 × 69 × 26.3 mm
Material	PETG, printed on a Bambu Lab H2D
Parts	Front shell, back plate, WAKE, User/Boot, RST, power slider
Versions	Four M2 screws, or a snap lock on all four sides that clicks home and pries open from one edge
Back	Plain, magnetic-mount ready: a sunk seat for an optional 55 × 45 × 0.75 mm stick-on magnetic ring (the kind sold for magnetic phone mounts), 0.68 mm of plate behind it
Finish	Soft rounded back edge, bevelled openings, a seam line and one clean control strip; optional engraving on the front only (frame line, six page marks)

- USB-C and Grove are enclosed right up to their connectors, so nothing opens into the case (the case is not weatherproof). USB-C sits in a bevelled pocket any cable latches into; Grove has a 14 mm grip pocket and a 45° funnel down to a 10.0 × 5.0 mm bore, so the plug finds its socket. The only other opening in the back is the speaker grille.
- Controls you can find by feel: WAKE, User/Boot and RST are one-piece buttons that snap in from outside and stay put. WAKE stands 1.2 mm proud and User/Boot 0.8 mm; RST sits nearly flush (0.1 mm) against accidental presses. The power slider has a thumb notch and can't be pushed in.
- The antenna connector is captured by a saddle on the back plate, so it can never work loose.
- Everything else goes in with the back open, placed by a map engraved inside the plate: GNSS, Wi-Fi, speaker, battery, over the magnetic ring if you fit one. The solar connector stays inside in this version. The plate comes off even with WAKE fitted.

GET STARTED

Flash it, print it, put it together.

FLASH CAIRN

Hold **User/Boot**, tap **RST**, release User/Boot, then write the full Cairn image at `0x0` with esptool or a browser-based ESP flasher.

Later updates install over Wi-Fi from Settings → System → Firmware.

Back to stock: Seeed's factory firmware can be re-flashed at any time with Seeed's own tools (see Seeed wiki).

FIRST RUN

- Settings → Radio: region and TX power, within your local limits.
- Settings → Connect → Bluetooth: set a PIN, pair the MeshCore app.
- Settings → Location → GPS: Eco to save battery; On for tracking.
- Radio → Advert to announce yourself.

AT A GLANCE

Radio	SX1262 LoRa, 863–928 MHz; RP-SMA via the Cairn case	Audio	ES8311 codec, 0.8 W speaker, alert tones
Compute	ESP32-S3R8, Wi-Fi, Bluetooth LE	Power	3000 mAh LiPo; USB-C 5 V (board solar input not brought out in this case)
Display	3.2" IPS 320 × 240, capacitive touch	Case	87.5 × 69 × 26.3 mm, PETG, screw or snap-fit, magnetic-mount ready
Position	L76K GNSS, ceramic patch antenna, compass view	Firmware	Cairn 106 on MeshCore v1.17.1; USB or Wi-Fi updates

CONTACT AND NEXT STEPS

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Firmware, source and releases: github.com/cvhviz/Cairn

I'd welcome Seeed's feedback on Cairn and the case, and I'm glad to help with testing, documentation or a community showcase.

ASSEMBLE THE CASE

- 1 Front shell face-down; fit the antenna connector and its nut.
- 2 Drop the power slider into its pocket.
- 3 Lower the board in, screen first — the power switch and the display find their places.
- 4 Optional: stick the magnetic ring into its seat in the plate. Place GNSS, Wi-Fi, speaker and battery where the plate's floor map shows; plug everything in, antenna too.
- 5 Back plate on: four screws, or press until it clicks.
- 6 Buttons go in last, from outside: push WAKE, User/Boot and RST into their holes until each clicks.

Included: print-ready STLs (screw and snap versions), Bambu Studio projects, renders, and the full parametric OpenSCAD source.

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